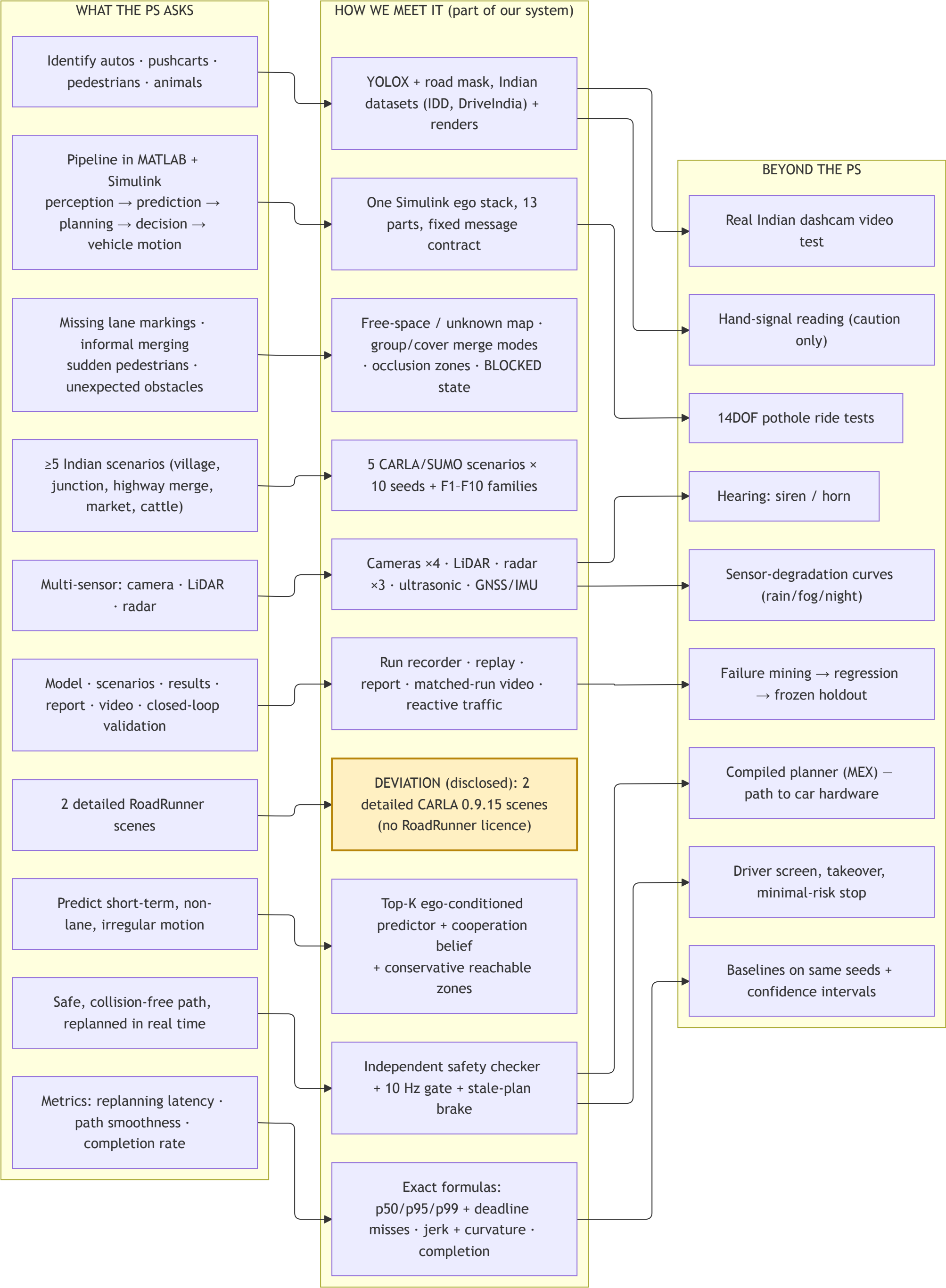


D6 · PS asks → how we meet it → beyond the PS

Gold = the one disclosed deviation (RoadRunner → CARLA). Status today: 1 meets · 22 partial · 15 not yet · 2 deviation (P14).



D6 · PS asks → how we meet it → beyond the PS — exact spec for redrawing

Redraw rule for Gemini / Nano Banana: keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

BOXES (grouped exactly as drawn):

■ WHAT THE PS ASKS

- [r1] Pipeline in MATLAB + Simulink perception → prediction → planning → decision → vehicle motion
- [r2] Multi-sensor: camera · LiDAR · radar
- [r3] Identify autos · pushcarts · pedestrians · animals
- [r4] Predict short-term, non-lane, irregular motion
- [r5] Safe, collision-free path, replanned in real time
- [r6] Missing lane markings · informal merging sudden pedestrians · unexpected obstacles
- [r7] ≥5 Indian scenarios (village, junction, highway merge, market, cattle)
- [r8] 2 detailed RoadRunner scenes
- [r9] Metrics: replanning latency · path smoothness · completion rate
- [r10] Model · scenarios · results · report · video · closed-loop validation

■ HOW WE MEET IT (part of our system)

- [o1] One Simulink ego stack, 13 parts, fixed message contract
- [o2] Cameras ×4 · LiDAR · radar ×3 · ultrasonic · GNSS/IMU
- [o3] YOLOX + road mask, Indian datasets (IDD, DriveIndia) + renders
- [o4] Top-K ego-conditioned predictor + cooperation belief + conservative reachable zones
- [o5] Independent safety checker + 10 Hz gate + stale-plan brake
- [o6] Free-space / unknown map · group/cover merge modes · occlusion zones · BLOCKED state
- [o7] 5 CARLA/SUMO scenarios × 10 seeds + F1-F10 families
- [o8] DEVIATION (disclosed): 2 detailed CARLA 0.9.15 scenes (no RoadRunner licence)
- [o9] Exact formulas: p50/p95/p99 + deadline misses · jerk + curvature · completion
- [o10] Run recorder · replay · report · matched-run video · reactive traffic

■ BEYOND THE PS

- [b1] Hearing: siren / horn
- [b2] Hand-signal reading (caution only)
- [b3] Real Indian dashcam video test
- [b4] Failure mining → regression → frozen holdout
- [b5] Sensor-degradation curves (rain/fog/night)
- [b6] 14DOF pothole ride tests
- [b7] Compiled planner (MEX) – path to car hardware
- [b8] Baselines on same seeds + confidence intervals
- [b9] Driver screen, takeover, minimal-risk stop

CONNECTIONS (19 arrows, from → to : what flows):

Pipeline in MATLAB + Simulink perception → prediction → planning → decision → vehicle motion → One Simulink ego stack, 13 parts, fixed message contract
Multi-sensor: camera · LiDAR · radar → Cameras ×4 · LiDAR · radar ×3 · ultrasonic · GNSS/IMU
Identify autos · pushcarts · pedestrians · animals → YOLOX + road mask, Indian datasets (IDD, DriveIndia) + renders
Predict short-term, non-lane, irregular motion → Top-K ego-conditioned predictor + cooperation belief + conservative reachable zones
Safe, collision-free path, replanned in real time → Independent safety checker + 10 Hz gate + stale-plan brake
Missing lane markings · informal merging sudden pedestrians · unexpected obstacles → Free-space / unknown map · group/cover merge modes · occlusion zones · BLOCKED state
≥5 Indian scenarios (village, junction, highway merge, market, cattle) → 5 CARLA/SUMO scenarios × 10 seeds + F1-F10 families
2 detailed RoadRunner scenes → DEVIATION (disclosed): 2 detailed CARLA 0.9.15 scenes (no RoadRunner licence)
Metrics: replanning latency · path smoothness · completion rate → Exact formulas: p50/p95/p99 + deadline misses · jerk + curvature · completion
Model · scenarios · results · report · video · closed-loop validation → Run recorder · replay · report · matched-run video · reactive traffic
YOLOX + road mask, Indian datasets (IDD, DriveIndia) + renders → Real Indian dashcam video test
Cameras ×4 · LiDAR · radar ×3 · ultrasonic · GNSS/IMU → Hearing: siren / horn
YOLOX + road mask, Indian datasets (IDD, DriveIndia) + renders → Hand-signal reading (caution only)
Run recorder · replay · report · matched-run video · reactive traffic → Failure mining → regression → frozen holdout
Cameras ×4 · LiDAR · radar ×3 · ultrasonic · GNSS/IMU → Sensor-degradation curves (rain/fog/night)
One Simulink ego stack, 13 parts, fixed message contract → 14DOF pothole ride tests
Independent safety checker + 10 Hz gate + stale-plan brake → Compiled planner (MEX) – path to car hardware
Exact formulas: p50/p95/p99 + deadline misses · jerk + curvature · completion → Baselines on same seeds + confidence intervals
Independent safety checker + 10 Hz gate + stale-plan brake → Driver screen, takeover, minimal-risk stop

Mermaid source (paste into mermaid.live to re-render)

```

flowchart LR
    subgraph PSR["WHAT THE PS ASKS"]
        r1["Pipeline in MATLAB + Simulink<br/>perception → prediction → planning → decision → vehicle motion"]
        r2["Multi-sensor: camera · LiDAR · radar"]
        r3["Identify autos · pushcarts · pedestrians · animals"]
        r4["Predict short-term, non-lane, irregular motion"]
        r5["Safe, collision-free path, replanned in real time"]
        r6["Missing lane markings · informal merging<br/>sudden pedestrians · unexpected obstacles"]
        r7["≥5 Indian scenarios (village, junction, highway merge, market, cattle)"]
        r8["2 detailed RoadRunner scenes"]
        r9["Metrics: replanning latency · path smoothness · completion rate"]
        r10["Model · scenarios · results · report · video · closed-loop validation"]
    end
    end
    subgraph OURS["HOW WE MEET IT (part of our system)"]
        o1["One Simulink ego stack, 13 parts, fixed message contract"]
        o2["Cameras x4 · LiDAR · radar x3 · ultrasonic · GNSS/IMU"]
        o3["YOLOX + road mask, Indian datasets (IDD, DriveIndia) + renders"]
        o4["Top-K ego-conditioned predictor + cooperation belief<br/>+ conservative reachable zones"]
        o5["Independent safety checker + 10 Hz gate + stale-plan brake"]
        o6["Free-space / unknown map · group/cover merge modes<br/>· occlusion zones · BLOCKED state"]
        o7["5 CARLA/SUMO scenarios × 10 seeds + F1-F10 families"]
        o8["DEVIATION (disclosed): 2 detailed CARLA 0.9.15 scenes<br/>(no RoadRunner licence)"]
        o9["Exact formulas: p50/p95/p99 + deadline misses · jerk + curvature · completion"]
        o10["Run recorder · replay · report · matched-run video · reactive traffic"]
    end
    end
    subgraph BEY["BEYOND THE PS"]
        b1["Hearing: siren / horn"]
        b2["Hand-signal reading (caution only)"]
        b3["Real Indian dashcam video test"]
        b4["Failure mining → regression → frozen holdout"]
        b5["Sensor-degradation curves (rain/fog/night)"]
        b6["14DOF pothole ride tests"]
        b7["Compiled planner (MEX) – path to car hardware"]
        b8["Baselines on same seeds + confidence intervals"]
        b9["Driver screen, takeover, minimal-risk stop"]
    end
    end
    r1 --> o1
    r2 --> o2
    r3 --> o3
    r4 --> o4
    r5 --> o5
    r6 --> o6
    r7 --> o7
    r8 --> o8
    r9 --> o9
    r10 --> o10
    o3 --> b3
    o2 --> b1
    o3 --> b2
    o10 --> b4
    o2 --> b5
    o1 --> b6
    o5 --> b7
    o9 --> b8
    o5 --> b9
    classDef dev fill:#fff0c2,stroke:#b8860b,stroke-width:2px
    class o8 dev

```